Quantum mechanics suggests that we're going to have to modify this story somewhat. When a spin is measured, the wave function branches via decoherence, a single world splits into two, and there are now two people where I used to be just one. It makes no sense to ask which one is "really me." Likewise, before the branching happens, it makes no sense to wonder which branch "I" will end up in. Both of them have every right to think of themselves as "me."

In a classical universe, identifying a single individual as a person aging through time is generally unproblematic. At any moment a person is a certain arrangement of atoms, but it's not the individual atoms that matter; to a large extent our atoms are replaced over time. What matters is the pattern that we form, and the continuity of that pattern, especially in the memories of the person under consideration.

The new feature of quantum mechanics is the duplication of that pattern when the wave function branches. That's no reason to panic. We just have to adjust our notion of personal identity through time to account for a situation that we never had reason to contemplate over the millennia of pre-scientific human evolution.

As stubborn as our identity is, the concept of a single person extending from birth to death was always just a useful approximation. The person you are right now is not exactly the same as the person you were a year ago, or even a second ago. Your atoms are in slightly different locations, and some of your atoms might have been exchanged for new ones. (If you're eating while reading, you might have more atoms now than you had a moment ago.) If we wanted to be more precise than usual, rather than talking about "you," we should talk about "you at 5:00 p.m.," "you at 5:01 p.m.," and so on.

The idea of a unified "you" is useful not because all of these different collections of atoms at different moments of time are literally the same, but because they are related to one another in an obvious way. They describe a real pattern. You at one moment descend from you at an earlier moment, through the evolution of the individual atoms within you and the possible addition or subtraction of a few of them. Philosophers have thought this through, of course; Derek Parfit, in particular, suggested that identity through time is a matter of one instance in your life "standing in Relation R" to another instance, where Relation R says that your future self shares psychological continuity with your past self.